Rational Expressions

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Simplifying radical expressions

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(5)$.

Final Answer: 21. Substitute $x = 5$: $f(5) = 3(5) + 6 = 21$.

2 [Foundational] • Target: Simplifying radical expressions

Find the positive solutions to the quadratic equation $x^2 - 13x + 36 = 0$.

Final Answer: $x = 4$ and $x = 9$. Factoring $(x - 4)(x - 9) = 0$ yields the roots $x = 4$ and $x = 9$.

3 [Foundational] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(2)$.

Final Answer: 15. Substitute $x = 2$: $f(2) = 4(2) + 7 = 15$.

4 [Foundational] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

Final Answer: $x = 1$ and $x = 3$. Factoring $(x - 1)(x - 3) = 0$ yields the roots $x = 1$ and $x = 3$.

5 [Foundational] • Target: Operations with radical expressions

A data set recorded for operations with radical expressions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 21.5. The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

6 [Foundational] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(3)$.

Final Answer: 19. Substitute $x = 3$: $f(3) = 4(3) + 7 = 19$.

7 [Foundational] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 6x + 8 = 0$.

Final Answer: $x = 2$ and $x = 4$. Factoring $(x - 2)(x - 4) = 0$ yields the roots $x = 2$ and $x = 4$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(4)$.

Final Answer: 23. Substitute $x = 4$: $f(4) = 4(4) + 7 = 23$.

9 [Intermediate] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

Final Answer: $x = 3$ and $x = 5$. Factoring $(x - 3)(x - 5) = 0$ yields the roots $x = 3$ and $x = 5$.

10 [Intermediate] • Target: Operations with radical expressions

For the model $f(x) = 4x + 7$ in operations with radical expressions, evaluate $f(5)$.

Final Answer: 27. Substitute $x = 5$: $f(5) = 4(5) + 7 = 27$.

11 [Intermediate] • Target: Operations with radical expressions

Find the positive solutions to the quadratic equation $x^2 - 10x + 24 = 0$.

Final Answer: $x = 4$ and $x = 6$. Factoring $(x - 4)(x - 6) = 0$ yields the roots $x = 4$ and $x = 6$.

12 [Intermediate] • Target: Graphs and characteristics of square root functions

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(2)$.

Final Answer: 18. Substitute $x = 2$: $f(2) = 5(2) + 8 = 18$.

13 [Intermediate] • Target: Graphs and characteristics of square root functions

A data set recorded for graphs and characteristics of square root functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

14 [Intermediate] • Target: Graphs and characteristics of square root functions

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(3)$.

Final Answer: 23. Substitute $x = 3$: $f(3) = 5(3) + 8 = 23$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Graphs and characteristics of square root functions • Cross-Lesson: 9.01 Trigonometric Functions

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(4)$.

Final Answer: 28. Substitute $x = 4$: $f(4) = 5(4) + 8 = 28$.

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16 [Challenging] • Target: Graphs and characteristics of square root functions • Cross-Lesson: 1.01 Functions & Their Graphs

For the model $f(x) = 5x + 8$ in graphs and characteristics of square root functions, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 5(5) + 8 = 33$.

17 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 10.01 Matrices & Systems

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(2)$.

Final Answer: 21. Substitute $x = 2$: $f(2) = 6(2) + 9 = 21$.

18 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 11.01 Probability & Statistics

A data set recorded for graphs and characteristics of cube root functions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 23.5. The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

19 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 2.01 Quadratics & Complex Numbers

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(3)$.

Final Answer: 27. Substitute $x = 3$: $f(3) = 6(3) + 9 = 27$.

20 [Challenging] • Target: Graphs and characteristics of cube root functions • Cross-Lesson: 3.01 Polynomial Functions

For the model $f(x) = 6x + 9$ in graphs and characteristics of cube root functions, evaluate $f(4)$.

Final Answer: 33. Substitute $x = 4$: $f(4) = 6(4) + 9 = 33$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(2)$.

a. 16 ✓

b. 22

c. 12

d. 20

Substitute $x = 2$: $f(2) = 6(2) + 4 = 16$.

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22 [Foundational] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 7x + 10 = 0$.

a. $x = 2$ and $x = 6$

b. $x = 2$ **and** $x = 5$ ✓

c. $x = -2$ and $x = -5$

d. $x = 7$ and $x = 10$

Factoring $(x - 2)(x - 5) = 0$ yields the roots $x = 2$ and $x = 5$.

23 [Foundational] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(5)$.

a. 34 ✓

b. 40

c. 30

d. 38

Substitute $x = 5$: $f(5) = 6(5) + 4 = 34$.

24 [Foundational] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(2)$.

a. 9 ✓

b. 11

c. 4

d. 13

Substitute $x = 2$: $f(2) = 2(2) + 5 = 9$.

25 [Foundational] • Target: Fractional powers

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.

a. $x = 2$ and $x = 7$

b. $x = 2$ **and** $x = 6$ ✓

c. $x = -2$ and $x = -6$

d. $x = 8$ and $x = 12$

Factoring $(x - 2)(x - 6) = 0$ yields the roots $x = 2$ and $x = 6$.

26 [Foundational] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(5)$.

a. 15 ✓

b. 17

c. 10

d. 19

Substitute $x = 5$: $f(5) = 2(5) + 5 = 15$.

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27 [Foundational] • Target: Simplifying radical expressions

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(2)$.

a. 12 ✓

b. 15

c. 6

d. 16

Substitute $x = 2$: $f(2) = 3(2) + 6 = 12$.

28 [Foundational] • Target: Simplifying radical expressions

Find the positive solutions to the quadratic equation $x^2 - 9x + 14 = 0$.

a. $x = 2$ and $x = 8$

b. $x = 2$ **and** $x = 7$ ✓

c. $x = -2$ and $x = -7$

d. $x = 9$ and $x = 14$

Factoring $(x - 2)(x - 7) = 0$ yields the roots $x = 2$ and $x = 7$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 5x + 4 = 0$.

a. $x = 1$ and $x = 5$

b. $x = 1$ **and** $x = 4$ ✓

c. $x = -1$ and $x = -4$

d. $x = 5$ and $x = 4$

Factoring $(x - 1)(x - 4) = 0$ yields the roots $x = 1$ and $x = 4$.

30 [Intermediate] • Target: Review: Laws of exponents

A data set recorded for review: laws of exponents consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean?

a. 16.5

b. 17.5

c. 18.5 ✓

d. 20.5

The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

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31 [Intermediate] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(3)$.

a. 22 ✓

b. 28

c. 18

d. 26

Substitute $x = 3$: $f(3) = 6(3) + 4 = 22$.

32 [Intermediate] • Target: Review: Laws of exponents

For the model $f(x) = 6x + 4$ in review: laws of exponents, evaluate $f(4)$.

a. 28 ✓

b. 34

c. 24

d. 32

Substitute $x = 4$: $f(4) = 6(4) + 4 = 28$.

33 [Intermediate] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 9x + 18 = 0$.

a. $x = 3$ and $x = 7$

b. $x = 3$ **and** $x = 6$ ✓

c. $x = -3$ and $x = -6$

d. $x = 9$ and $x = 18$

Factoring $(x - 3)(x - 6) = 0$ yields the roots $x = 3$ and $x = 6$.

34 [Intermediate] • Target: Review: Laws of exponents

Find the positive solutions to the quadratic equation $x^2 - 11x + 28 = 0$.

a. $x = 4$ and $x = 8$

b. $x = 4$ **and** $x = 7$ ✓

c. $x = -4$ and $x = -7$

d. $x = 11$ and $x = 28$

Factoring $(x - 4)(x - 7) = 0$ yields the roots $x = 4$ and $x = 7$.

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35 [Intermediate] • Target: Fractional powers

A data set recorded for fractional powers consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

36 [Intermediate] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(3)$.

a. 11 ✓

b. 13

c. 6

d. 15

Substitute $x = 3$: $f(3) = 2(3) + 5 = 11$.

37 [Intermediate] • Target: Fractional powers

For the model $f(x) = 2x + 5$ in fractional powers, evaluate $f(4)$.

a. 13 ✓

b. 15

c. 8

d. 17

Substitute $x = 4$: $f(4) = 2(4) + 5 = 13$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Fractional powers • Cross-Lesson: 11.01 Probability & Statistics

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

a. $x = 3$ and $x = 8$

b. $x = 3$ **and** $x = 7$ ✓

c. $x = -3$ and $x = -7$

d. $x = 10$ and $x = 21$

Factoring $(x - 3)(x - 7) = 0$ yields the roots $x = 3$ and $x = 7$.

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39 [Challenging] • Target: Fractional powers • Cross-Lesson: 2.01 Quadratics & Complex Numbers

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

a. $x = 4$ and $x = 9$

b. $x = 4$ **and** $x = 8$ ✓

c. $x = -4$ and $x = -8$

d. $x = 12$ and $x = 32$

Factoring $(x - 4)(x - 8) = 0$ yields the roots $x = 4$ and $x = 8$.

40 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 3.01 Polynomial Functions

Find the positive solutions to the quadratic equation $x^2 - 7x + 6 = 0$.

a. $x = 1$ and $x = 7$

b. $x = 1$ **and** $x = 6$ ✓

c. $x = -1$ and $x = -6$

d. $x = 7$ and $x = 6$

Factoring $(x - 1)(x - 6) = 0$ yields the roots $x = 1$ and $x = 6$.

41 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 5.01 Exponential & Logarithmic Functions

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(3)$.

a. 15 ✓

b. 18

c. 9

d. 19

Substitute $x = 3$: $f(3) = 3(3) + 6 = 15$.

42 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 6.01 Radicals & Rational Exponents

For the model $f(x) = 3x + 6$ in simplifying radical expressions, evaluate $f(4)$.

a. 18 ✓

b. 21

c. 12

d. 22

Substitute $x = 4$: $f(4) = 3(4) + 6 = 18$.

43 [Challenging] • Target: Simplifying radical expressions • Cross-Lesson: 7.01 Sequences & Series

Find the positive solutions to the quadratic equation $x^2 - 11x + 24 = 0$.

a. $x = 3$ and $x = 9$

b. $x = 3$ **and** $x = 8$ ✓

c. $x = -3$ and $x = -8$

d. $x = 11$ and $x = 24$

Factoring $(x - 3)(x - 8) = 0$ yields the roots $x = 3$ and $x = 8$.

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44 [Challenging] • Target: Operations with radical expressions • Cross-Lesson: 8.01 Conic Sections

A data set recorded for operations with radical expressions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

45 [Challenging] • Target: Graphs and characteristics of square root functions • Cross-Lesson: 9.01 Trigonometric Functions

A data set recorded for graphs and characteristics of square root functions consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean?

a. 20.5

b. 21.5

c. 22.5 ✓

d. 24.5

The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

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